Digital Twin and Warehouse Simulation for Autonomous Navigation

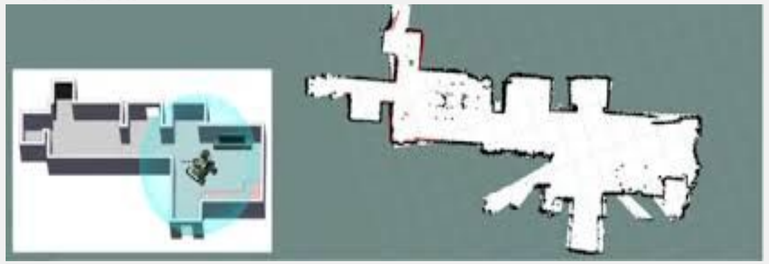
Gazebo + AWS RoboMaker small warehouse world + ROS navigation stack

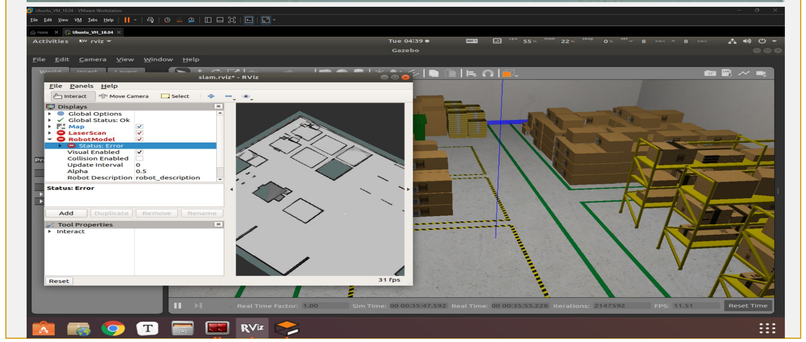
Problem

Autonomous warehouse robots needs reliable localization, path planning, and obstacle avoidance before they ever reach the shop floor. Developing those capabilities only on hardware is slow and hard to reproduce. This project creates a ROSMASTER X3 digital twin inside the AWS RoboMaker small warehouse world so navigation can be built, tuned, and debugged in simulation first. The result is a repeatable workflow that connects realistic sensors, a structured warehouse map, and a full ROS navigation pipeline.

Goals and Contributions

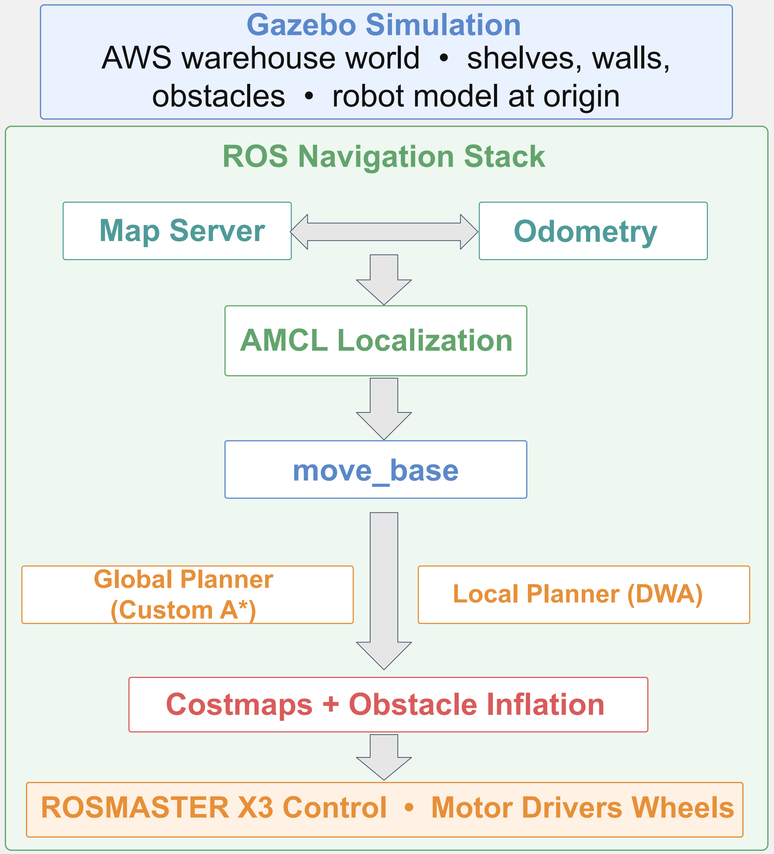
- **Build a Gazebo digital twin of the ROSMASTER X3 with LiDAR, RGB camera, odometry, and mecanum-wheel motion.**
- **Integrate the AWS small warehouse world as a repeatable indoor testbed.**
- **Support autonomous navigation with AMCL, move_base, DWA local planning, and a custom A* global planner.**
- **Enable both pre-mapped navigation and SLAM-based map creation.**





Methodology / System Architecture

Gazebo publishes sensor streams, the navigation stack localizes and plans, and the controller returns /cmd_vel commands to the robot model.



Operation Modes

Mode A: Pre-mapped Navigation

Launch navigation_aws_warehouse.launch

Load the saved map with AMCL.

- **Set a 2D Nav Goal in RViz and follow the planned path.**

Mode B: SLAM Mapping

Run the SLAM or Cartographer launch file.

Teleoperate the robot to scan the warehouse.

- **Save new .pgm and .yaml map files.**

Workflow

1. **Launch Simulation**
   Gazebo loads warehouse world and spawns robot with active sensors.
2. **Localization**
   AMCL fuses LiDAR + odometry to estimate pose on the map.
3. **Set Navigation Goal**
   User sets target in RViz, A* planner computes optimal path.
4. **Path Execution**
   DWA local planner follows path while avoiding obstacles.
5. **Goal Achievement**
   Robot reaches target, ready for next command.

Key Specifications

Project Outcome and Future Work

- **Gazebo digital twin and AWS warehouse world run together as one navigation testbed.**
- **RViz goals drive autonomous motion on a saved warehouse map.**
- **Custom A* global planning works alongside DWA local obstacle avoidance.**
- **SLAM launches support building and saving new occupancy maps.**

Next steps

- **Validate the same stack on the physical ROSMASTER X3 platform.**
- **Add dynamic-obstacle benchmarks and warehouse-task scenarios.**
- **Expand camera-based perception for inspection and inventory work.**

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